AIPIL (COSMOPOLITAN) – YIN CONSORTIUM
INNOVATION DEVELOPMENT AND EFFECTIVENESS IN THE ACQUISITION OF SKILLS (IDEAS)
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Capstone Project Guidelines for Participants

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> *A capstone project is the participant's opportunity to demonstrate that they can apply the knowledge, technical skills, professional behaviours, and ethical judgment developed during the IDEAS programme to a meaningful real-world problem.*

1. Purpose of the capstone

The capstone is intended to:

- demonstrate practical competence in the participant's selected IDEAS skill track;
- solve a clearly defined problem affecting a person, organisation, business, or community;
- produce evidence that can be reviewed by trainers, assessors, internal quality assurance personnel, and—where applicable—external quality assurance personnel;
- strengthen employability, entrepreneurship, freelancing, internship, and industry-placement prospects;
- encourage teamwork, communication, creativity, critical thinking, ethical technology use, and a growth mindset; and
- contribute to the IDEAS Project's wider goal of developing future-ready digital skills and solutions with social and economic value.

The capstone should be treated as both a **real product or service prototype** and an **assessment portfolio**.

2. Who should complete a capstone

Every participant who has completed the relevant training units should submit a capstone, either:

- individually; or
- as part of an approved team.

A team submission must still make each participant's individual contribution clear. A team may share a product, but no participant will receive credit for work they cannot explain, demonstrate, or evidence.

3. Suitable project areas

Projects should relate directly to the participant's trained track. The current project materials identify the following broad areas:

1. **Artificial Intelligence** — including AI, machine learning, data science, business intelligence, robotics, prompt engineering, or an applied AI solution.
2. **Advanced Software Development and Engineering** — including frontend, backend, full-stack, mobile, web, or other approved software-development work.

3 **Advanced Creative Skills and Digital Marketing** — including graphic design, branding, campaign design, content systems, social-media marketing, or digital engagement.

4 **Enterprise Digital Skills** — including practical Google Workspace or Microsoft 365 solutions for small and medium-sized organisations.

5 **Emerging Technology / Drone Technology** — where the participant has been trained and has access to appropriate equipment, supervision, and a safe operating environment.

A project may combine tracks—for example, a software application with an AI feature or a digital-marketing campaign supported by a data dashboard—but the participant must identify the primary track and the competencies being demonstrated.

4. What makes a strong capstone

A strong capstone should be:

- **Relevant:** addresses a genuine user, business, institutional, or community need.
- **Specific:** has a focused problem statement, target users, and defined scope.
- **Feasible:** can be completed with the time, skills, tools, budget, and equipment available.
- **Practical:** produces a working prototype, demonstrable service, campaign, portfolio, or tested solution—not only a written idea.
- **Inclusive:** considers accessibility, gender, disability, language, device availability, connectivity, affordability, and other barriers relevant to intended users.
- **Ethical:** respects privacy, intellectual property, consent, data protection, safety, and responsible technology practice.
- **Tested:** includes user feedback, functional testing, design review, or another credible form of validation.
- **Documented:** includes enough evidence for another person to understand, reproduce, operate, or assess the work.
- **Presentable:** can be explained clearly to a non-specialist stakeholder and defended in an oral/practical assessment.

5. Required project stages

Stage 1 — Identify and define the problem

Submit a short concept note containing:

- project title;
- participant name(s), cohort, and primary skill track;
- problem statement;
- intended users or beneficiaries;
- why the problem matters;
- proposed solution;
- expected benefit or outcome;
- relevant Sustainable Development Goal, community, industry, or institutional connection, if applicable; and
- initial list of tools, data, equipment, or external support required.

The problem must be stated in practical terms. Avoid broad topics such as “solve unemployment” or “improve education” without identifying a specific user problem that the project can address.

Stage 2 — Plan the solution

Prepare a project plan that identifies:

- objectives and success measures;
- key features or deliverables;
- work breakdown and milestones;
- roles and responsibilities for team members;
- risks, dependencies, and mitigation actions;
- accessibility and inclusion considerations;
- ethical, privacy, safety, and intellectual-property considerations; and
- the planned testing or validation method.

The trainer or facilitator will approve the scope before substantial implementation begins.

Stage 3 — Build or produce the solution

Develop the minimum viable solution first. Depending on the track, this may be:

- a functional website, application, dashboard, automation, or software prototype;
- an AI model, prompt workflow, data product, or decision-support prototype;
- a branding package, campaign, content system, or digital-marketing solution;
- an accessible productivity/workflow system using Google Workspace or Microsoft 365; or
- a safe, documented drone or emerging-technology demonstration.

Use version control, dated working files, design iterations, or another method that shows how the work developed.

Stage 4 — Test, improve, and validate

Testing should be proportionate to the project. Evidence may include:

- functional or usability tests;
- test cases and results;
- user interviews, surveys, or feedback forms;
- accessibility checks;
- campaign analytics or engagement results;
- model-performance measures and limitations;
- peer or mentor review;
- before-and-after comparisons; or
- a demonstration against defined acceptance criteria.

Record the feedback received, the changes made, and any limitations that remain.

Stage 5 — Present and defend the project

Each participant must be able to:

- explain the problem and intended users;
- demonstrate the solution or present the completed creative/service output;
- explain their individual contribution;
- justify major technical, design, and business decisions;
- describe testing and respond to questions about results;
- identify ethical, privacy, security, accessibility, and safety issues; and
- discuss next steps for improvement, deployment, employment, entrepreneurship, or industry use.

6. Minimum submission package

Each participant or team should submit the following, using the format and naming convention announced by the project team:

6 **Project concept note** — approved problem and proposed solution.

7 **Project plan** — objectives, milestones, roles, risks, and success measures.

8 **Final product or prototype** — live link, repository, demo file, design package, campaign assets, workflow, dataset/model, or other appropriate output.

9 **Technical or production documentation** — setup/use instructions, architecture or design explanation, tools used, and known limitations.

10 **Testing and validation evidence** — test results, user feedback, analytics, or review record.

11 **Individual contribution statement** — what each participant personally did and learned.

12 **Presentation/demo** — slides or other visual support, together with a live or recorded demonstration where appropriate.

13 **Reflective statement** — lessons learned, challenges, ethical considerations, and proposed next steps.

14 **Sources and attribution list** — datasets, code libraries, images, fonts, models, APIs, references, collaborators, and other third-party materials.

15 **Authenticity declaration** — confirmation that the submission is the participant's own work, with all assistance and third-party materials disclosed.

7. Track-specific evidence expectations

Artificial Intelligence

Include, where relevant:

- the data source, data permissions, and data-preparation method;
- the chosen model, prompt design, workflow, or algorithm;
- evaluation criteria and results;
- limitations, bias, hallucination/error risks, and human oversight;
- privacy and security controls; and
- a reproducible demonstration or documented usage procedure.

Software Development and Engineering

Include, where relevant:

- requirements and user stories;
- system design or architecture;
- readable, organised code;
- version history or repository evidence;
- testing and error handling;
- accessibility and responsive behaviour;
- deployment or installation instructions; and
- security and data-handling considerations.

Creative Skills and Digital Marketing

Include, where relevant:

- a defined audience and campaign objective;
- brand or design rationale;
- original editable assets and final exports;
- content calendar or campaign plan;
- platform choices and accessibility considerations;
- evidence of testing, feedback, reach, engagement, or conversion where available; and
- disclosure of stock, licensed, AI-generated, or third-party assets.

Enterprise Digital Skills

Include, where relevant:

- the organisation's workflow problem;
- the designed spreadsheet, form, dashboard, document system, automation, or collaboration setup;
- user roles and permissions;
- data-validation and error-prevention steps;
- a short user guide;
- evidence of usability and efficiency improvement; and
- privacy and access-control considerations.

Drone or Emerging Technology

Include, where relevant:

- the use case and operating environment;
- equipment and software used;
- risk assessment and safety controls;
- permissions and applicable regulatory considerations;
- flight or demonstration log;
- captured outputs and data-handling arrangements; and
- evidence that the participant can explain the system and operate it safely.

8. Assessment principles and suggested weighting

Please find below the rubric that will be used for lead review:

Criterion	Suggested weighting
Problem definition, relevance, and user understanding	15%
Technical or creative quality of the solution	25%
Application of track competencies	20%
Testing, evidence, and iteration	10%
Inclusion, accessibility, ethics, privacy, and safety	10%
Documentation and individual contribution	10%
Presentation, demonstration, and oral defence	10%
Total	**100%**

A participant should not pass solely because the final product looks attractive. The assessor will confirm the participant's underlying knowledge, practical skill, reasoning, and ability to explain the work.

9. Assessment and quality-assurance requirements

The capstone will form part of the participant's assessment portfolio and will be handled consistently with the competency-assessment process:

- maintain candidate and project records;
- record the planned and actual assessment dates;
- identify the trainer/facilitator, assessor(s), and internal verifier/IQAM;
- use clear assessment criteria and record decisions;
- require a learner declaration of authenticity;
- conduct an oral question-and-answer component covering ethics, responsibilities, tools, workflows, and practical implementation;
- record whether each required unit or competency is achieved; and
- retain evidence for internal and external quality-assurance sampling.

Where a team project is submitted, assessors will ask individual questions and may require an individual practical task to verify authorship and competence.

10. Academic integrity, ethics, and responsible use of tools

Participants must:

- submit work that they created or materially contributed to;
- cite and attribute all external sources and collaborators;
- obtain consent before collecting personal data, images, recordings, or user feedback;
- avoid exposing passwords, private records, confidential business data, or personally identifiable information;

- disclose the use of generative AI, templates, code libraries, stock assets, datasets, and other third-party materials;
- verify AI-generated content and remain accountable for the final result;
- avoid plagiarism, impersonation, fabricated results, falsified user feedback, and unsupported claims;
- respect software licences, copyright, trademarks, and data-use restrictions; and
- follow all safety and regulatory requirements for physical equipment, drones, financial services, health-related tools, and other sensitive applications.

A participant may use AI or other digital tools as a learning or production aid only where the use is transparent and the participant can explain, test, and take responsibility for the submitted work.

11. Inclusion and reasonable support

The project will provide reasonable support so that participants—including women, people with disabilities, and participants with learning, language, sensory, or physical barriers—can demonstrate competence fairly. Participants should inform the project team early about support needs.

Possible adjustments include accessible file formats, additional explanation of instructions, guidance on assistive technology, alternative presentation methods, extended time where justified, accessible venues, and approved team roles. Adjustments must preserve the competency being assessed.

12. Recommended timeline

The project team will publish a dated schedule. A practical sequence is:

16 **Orientation and topic selection** — capstone briefing and examples.

17 **Concept approval** — problem statement and track mapping.

18 **Design review** — project plan, risks, ethics, and scope.

19 **Midpoint check** — working prototype or first complete draft.

20 **Testing and improvement** — evidence collection and iteration.

21 **Final submission** — portfolio and product/prototype.

22 **Demonstration and oral defence** — individual verification.

23 **Assessment and quality assurance** — assessor decision, sampling, feedback, and resubmission process if applicable.

24 **Showcase and progression** — exhibition, employer/industry review, internship, job, or entrepreneurship referrals where available.

13. Participant checklist before submission

- ☐My project is linked to an IDEAS skill track.
- ☐I have identified a specific user or beneficiary and a real problem.
- ☐My scope and success measures were approved.
- ☐I can demonstrate a working or reviewable output.
- ☐I have recorded my own contribution.
- ☐I have tested the work and acted on feedback.
- ☐I have documented limitations and next steps.
- ☐I have addressed accessibility, privacy, ethics, safety, and intellectual property.
- ☐I have cited external sources and disclosed tools, collaborators, and AI use.
- ☐I can explain and defend the work without relying on another team member.
- ☐I have included the authenticity declaration and all required evidence